



— MAMA - SYNTH · MICCAI 2026

Why we're working to move **beyond** **gadolinium**

59 million doses are injected every year. The science tells a bigger story than most people realise. →

— FIRST, THE HONEST PART

Gadolinium is a remarkable tool that saves lives every day.

For most patients, it's safe and clinically invaluable. We're not here to alarm anyone.

But the **aggregate picture**, across patient safety, experience, environment and sustainability, makes a compelling case for a contrast-free alternative.



Kidney patients are at real risk

In patients with impaired renal function, gadolinium can trigger **nephrogenic systemic fibrosis**, a progressive, sometimes fatal hardening of skin and organs.

173 documented cases · onset 2 days to 3 years



Not for everyone, and not non-invasive

Gadolinium can **cross the placenta** into fetal circulation, so radiology guidelines recommend avoiding routine use in pregnancy unless considered critical. And every scan needs an IV line, where contrast can leak into surrounding tissue (extravasation), occasionally causing injury.

Extravasation rate 0.045% of 1.1M injections



Allergic reactions, and the numbers are real

In an 8-year cohort of **154,539 patients**, acute allergic-like hypersensitivity reactions occurred in **0.4% of administrations**. And in patients who have reacted before, a separate study found reactions recur in about **30%** on re-exposure.

0.4% acute HSR rate · ~30% recurrence if prior reaction



It deposits, and lingers in the body

Gadolinium is retained in the brain and body for months to years, even in patients with **normal kidney function**. In December 2017 the FDA required a **new class-wide warning** for all GBCAs. The agency notes retention isn't proven to cause harm in healthy kidneys; some patients nonetheless report persistent symptoms, studied under the contested term "Gadolinium Deposition Disease."

FDA class-wide warning · December 2017



Contrast makes a long scan longer

A full dynamic contrast-enhanced breast MRI needs a pre-contrast scan plus four to six post-contrast sequences, typically a **30 to 60 minute** appointment. Remove the contrast dynamics and acquisition can fall to **~3 minutes**. And when patients describe what distresses them most about breast MRI, the **long duration** and feeling trapped rank at the top.

Full DCE ~30-60 min • vs ~3 min abbreviated



The injection is a barrier in itself

Offered a free contrast-enhanced screening MRI, only **58% of eligible high-risk women agreed**. Among the 512 who declined: 25.4% cited claustrophobia, **5.7% did not want the intravenous injection**, and 1.4% cited gadolinium-related risks or allergies. The needle and the agent itself keep women away from a test that can catch cancer early.

Only 58% agreed to contrast MRI (ACRIN 6666)

— IT DOESN'T STOP AT THE PATIENT

Then it enters the environment.

GBCAs pass through wastewater treatment **without breaking down**. So where do they go? →



It's in our drinking water

Because GBCAs aren't removed by standard treatment, they flow into rivers, groundwater and tap water worldwide. First detected in Germany in 1996, now confirmed on **every inhabited continent**.

Detected worldwide since 1996

Bau & Dulski, **Earth Planet. Sci. Lett.** 1996 · Schmidt et al., **Sci. Total Environ.** 2019



It's even in our beverages

A 2025 study detected gadolinium in **30 commercial beers** from Europe, Asia and Peru, plus energy drinks. An earlier study found it in cola served at fast-food chains across six German cities.

Gd anomaly 4-9× natural levels in every beer tested



It's accumulating in marine life

In controlled studies, mussels exposed to gadolinium showed **suppressed metabolism and cellular damage**, and sea urchin larvae suffered **disrupted skeletal development**. The likely mechanism: Gd^{3+} can mimic calcium (Ca^{2+}) and interfere with the calcium signalling that muscles, hearts and nerves depend on.

Mussel uptake $>2.5 \mu\text{g/g}$ · sea urchin development disrupted

Andrade et al., **Environ. Sci. Pollut. Res.** 2023 · Martino et al., **Aquatic Toxicology** 2018 · Moreira et al., **Environ. Pollution** 2025



A rare earth with a heavy footprint

Producing rare earth elements is chemically intensive, with ores roasted in **~98% sulphuric acid at ~500°C** and refined through repeated solvent extraction. The oxide group containing gadolinium (with samarium and europium) carries the **highest environmental footprint** of all rare earth groups: 1,074 MJ of energy and 55.6 kg CO₂-equivalent per kilogram produced. Meanwhile, an estimated **19 tons** of gadolinium enter EU waters each year (21 tons in the US).

55.6 kg CO₂-e/kg · 19 tons/yr into EU waters

— THE QUESTION WE'RE ASKING

What if we didn't need the injection at all?

MAMA-SYNTH is an international AI challenge at MICCAI 2026: can generative models synthesise diagnostic-quality post-contrast breast MRI from **pre-contrast images alone?**



Gd

Do you agree this is worth solving?

And what reasons have we missed? Tell us in the comments, we'd genuinely like to learn from this community.

ub.edu/mama-synth

MICCAI 2026

Test Phase opens June 25 · Results Release in August · Winners Announcement @ MICCAI 2026